

AUGMENTED REALITY

PENYEDERHANAAN GAMBAR

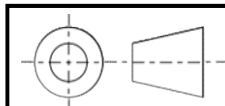
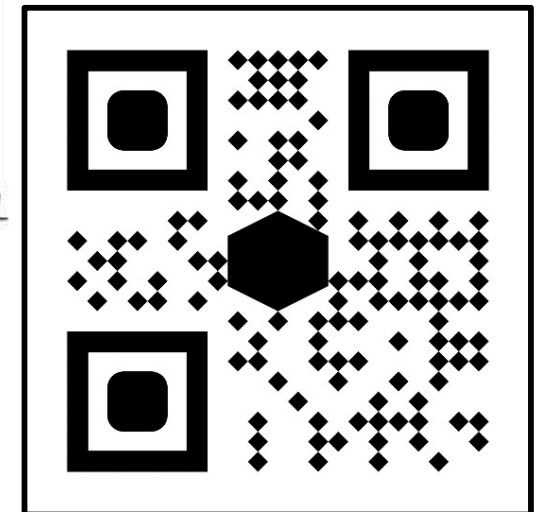
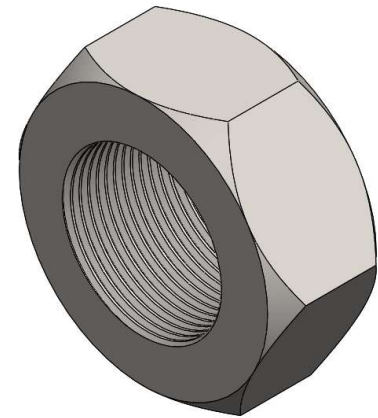
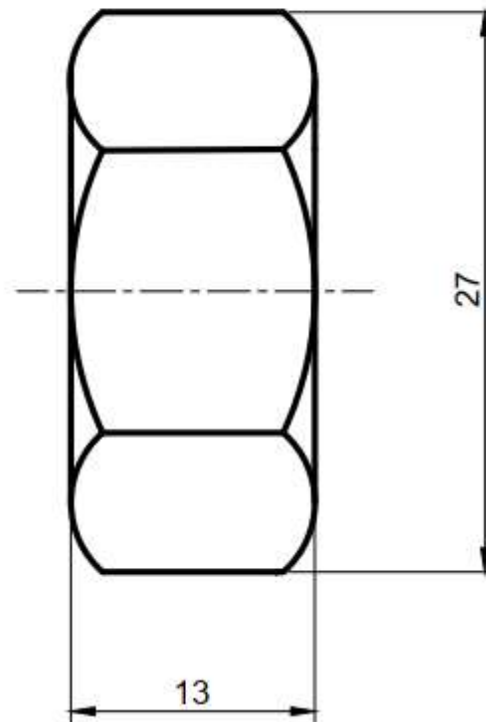
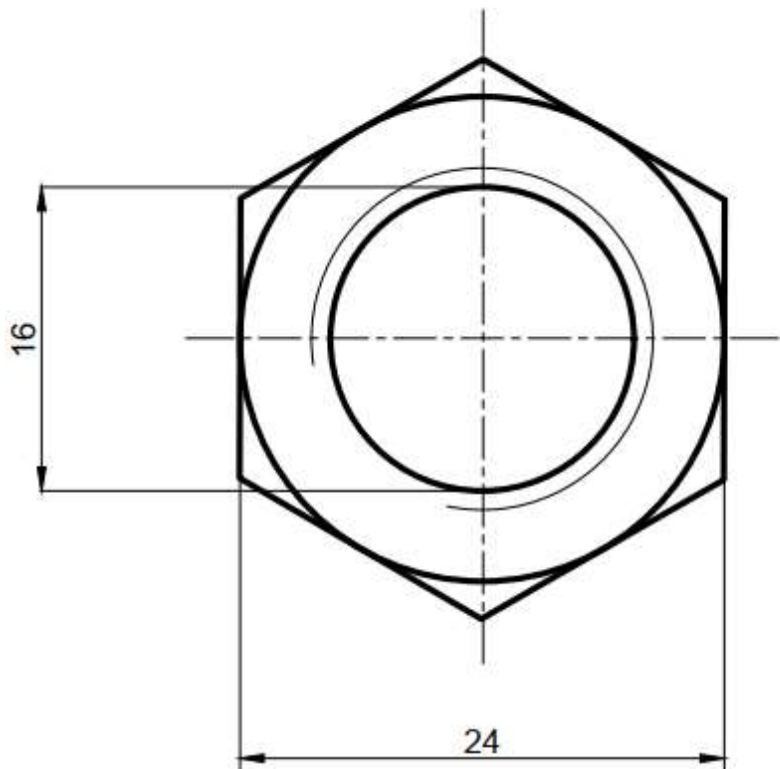
ARPeGa (Augmented Penyederhanaan Gambar) merupakan Aplikasi Augmented Reality yang dikembangkan sebagai modul praktikum Penyederhanaan Gambar (*simplified representation*) pada Mata Kuliah Menggambar Teknik (Teknik Mesin).

Setiap Lembar ARPeGa ini berisi:

1. Gambar Teknik 2 Dimensi
2. Gambar model 3D
3. Marker yang dipindai (*scan*) pada aplikasi ARPeGa

Adapun Gambar Teknik 2 Dimensi pada dokumen ini dibuat dengan ketentuan sebagai berikut:

- Dibuat tanpa skala
- Menekankan pada penyederhanaan gambar
- Satuan yang digunakan adalah milimeter
- Menggunakan sistem proyeksi sudut ketiga



Mur (Nut)

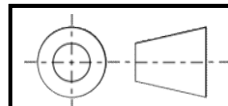
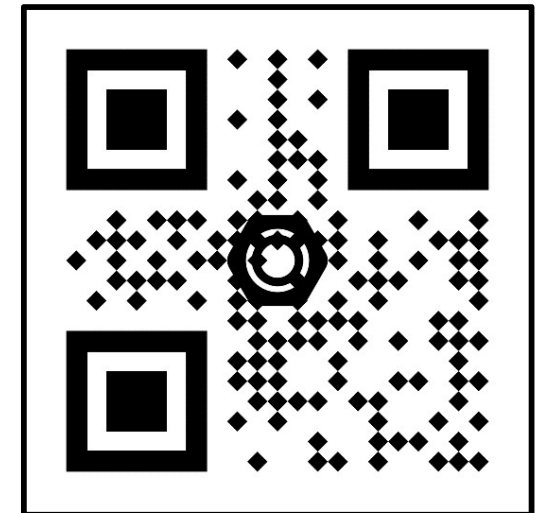
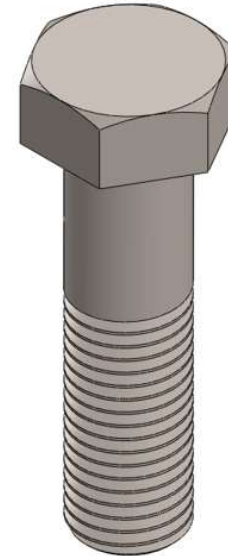
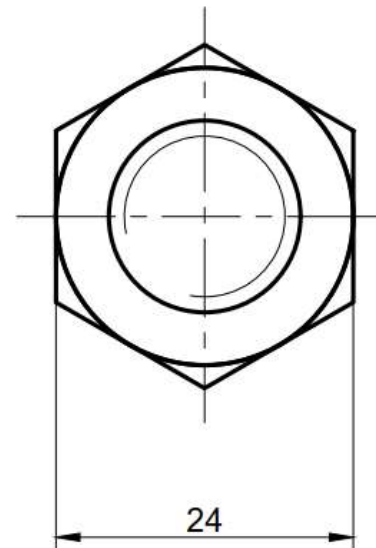
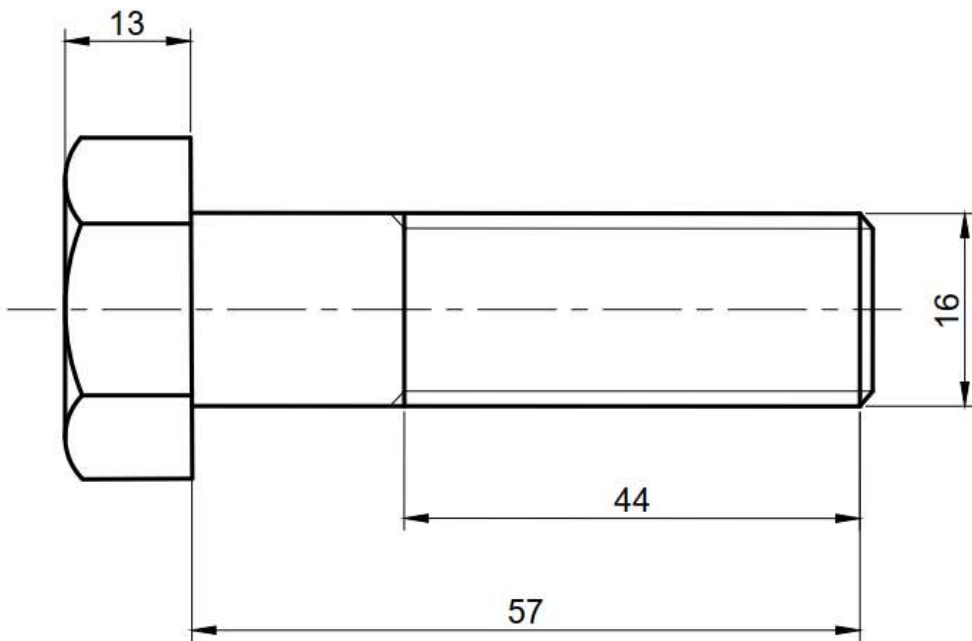
Tanpa skala

Satuan: mm



AR – Penyederhanaan Gambar

1



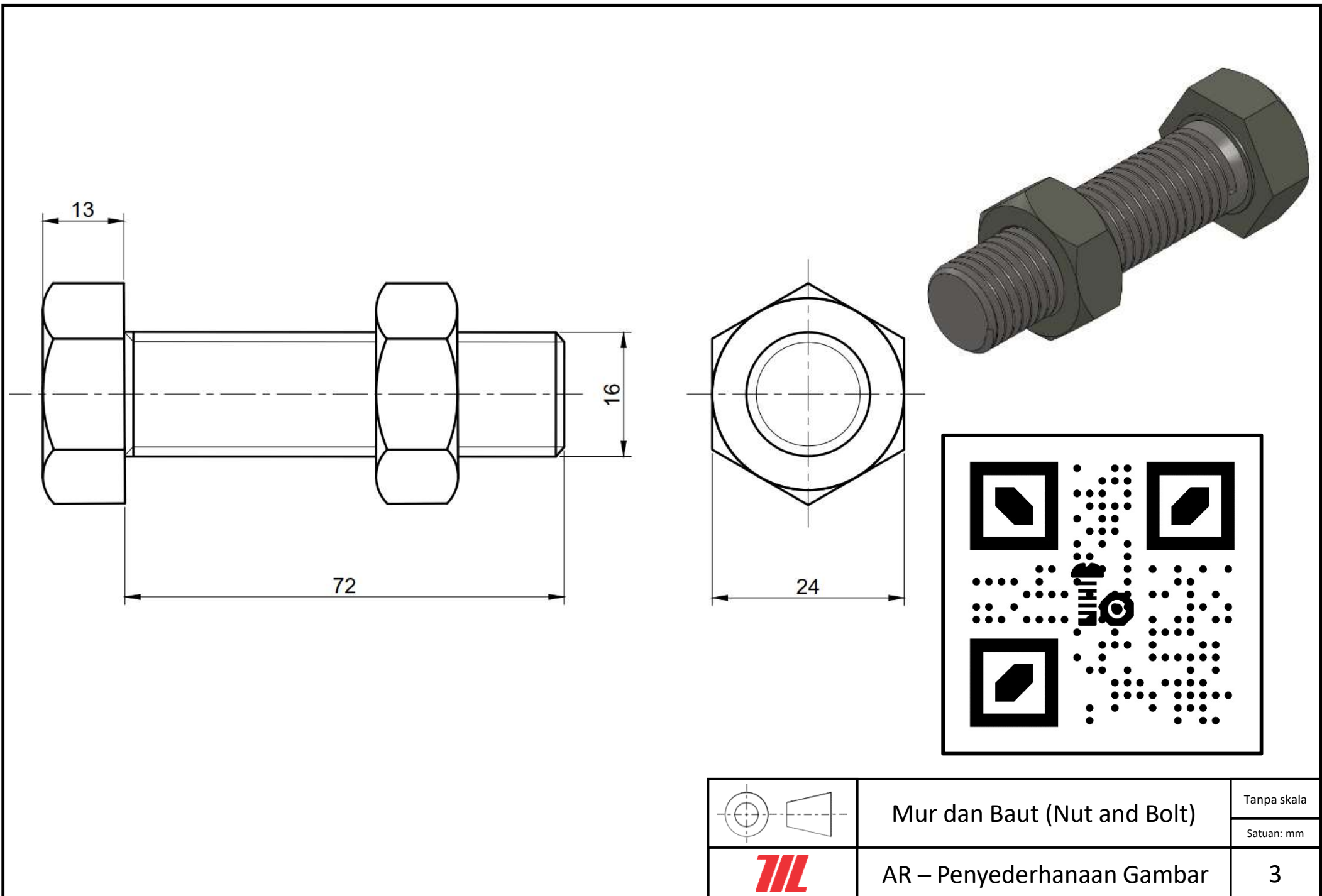
Baut (Bolt)

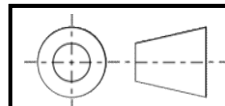
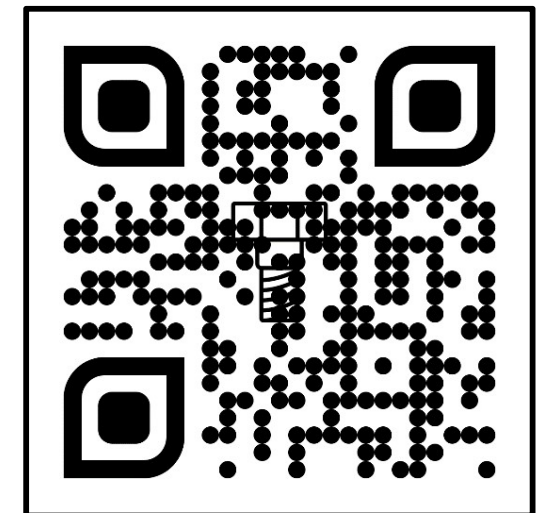
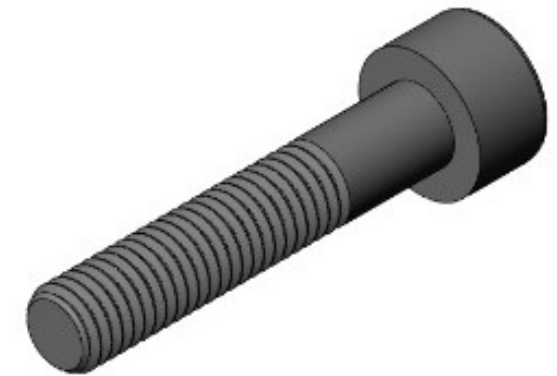
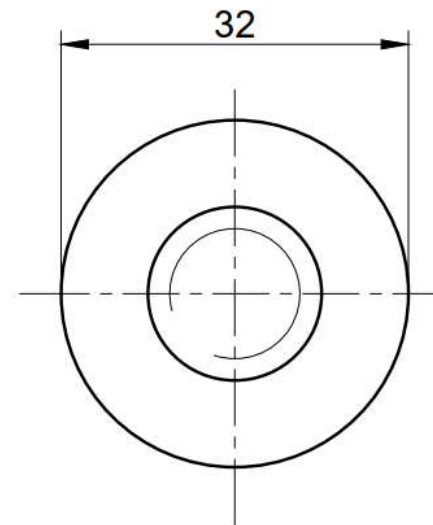
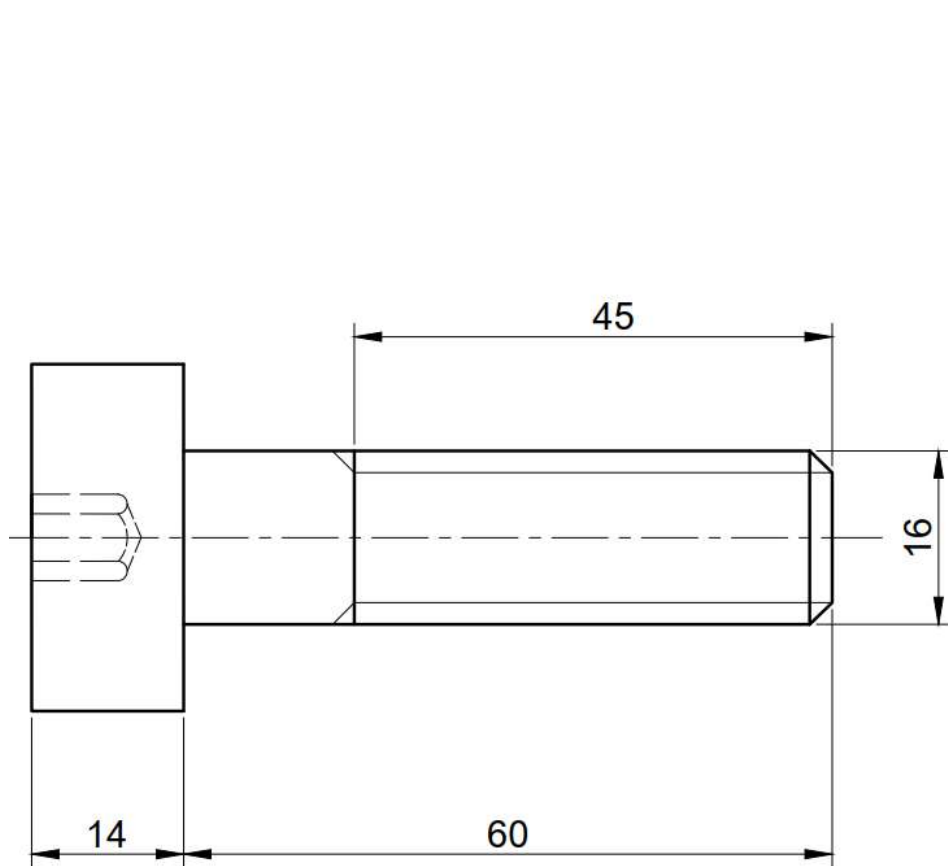
Tanpa skala

Satuan: mm

AR – Penyederhanaan Gambar

2





Baut (Counterbore head screw)

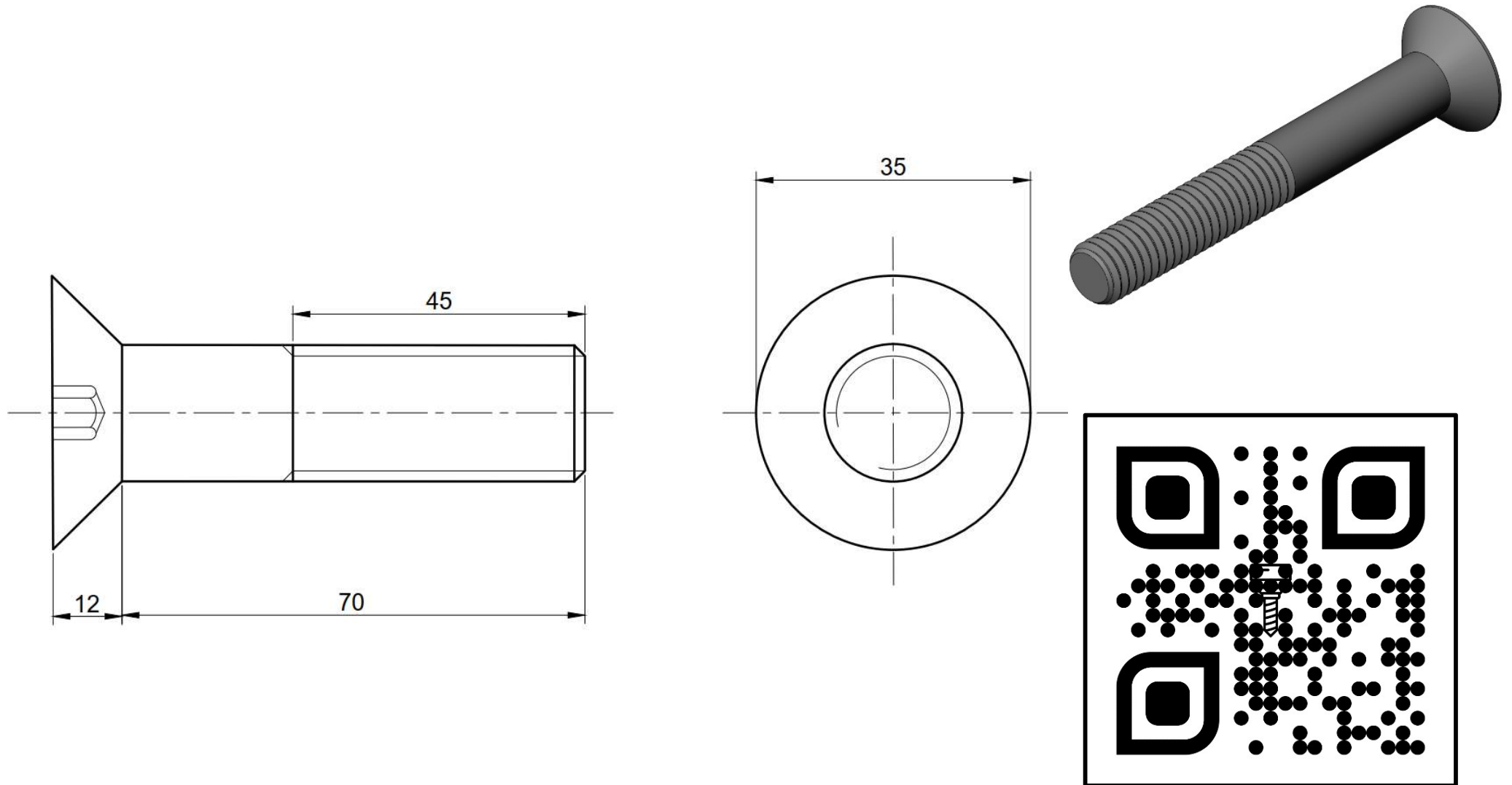
Tanpa skala

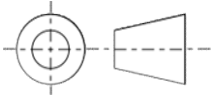
Satuan: mm

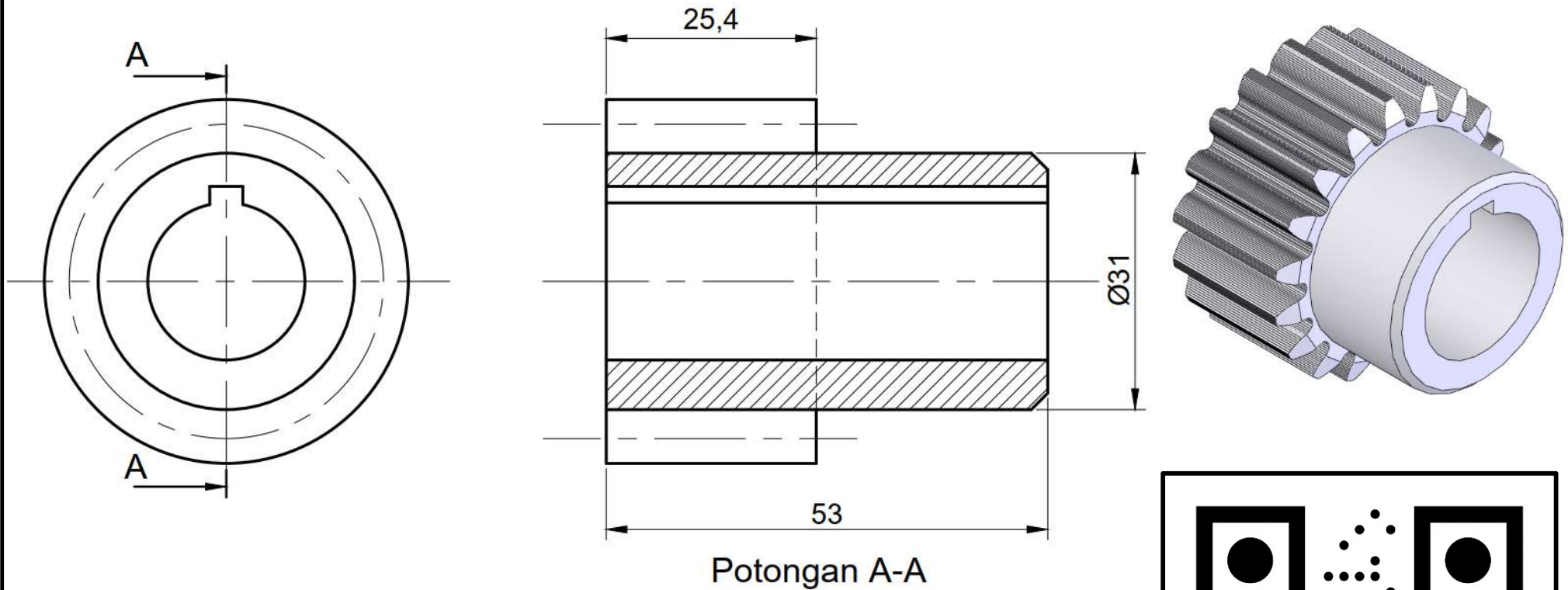


AR – Penyederhanaan Gambar

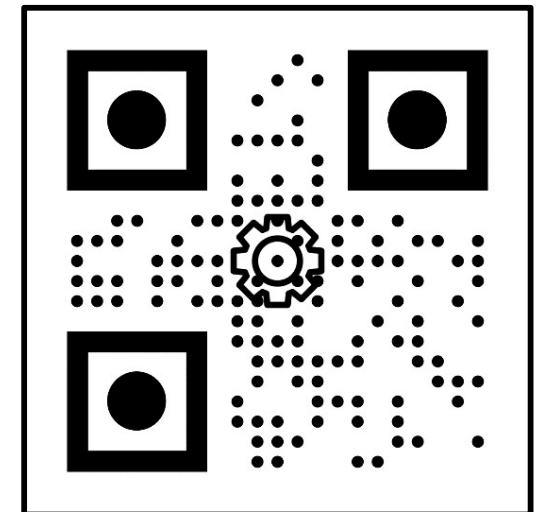
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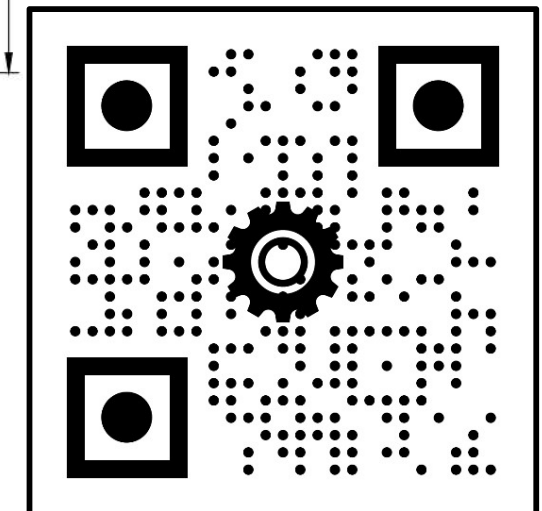
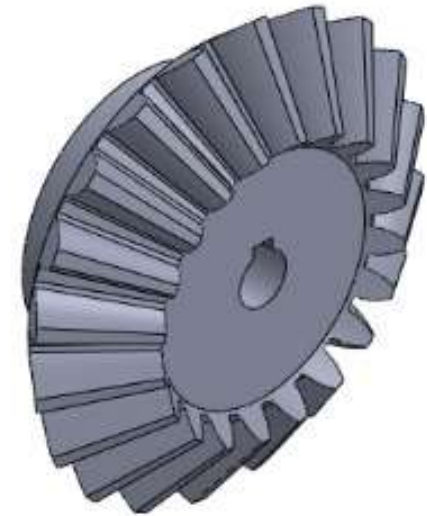
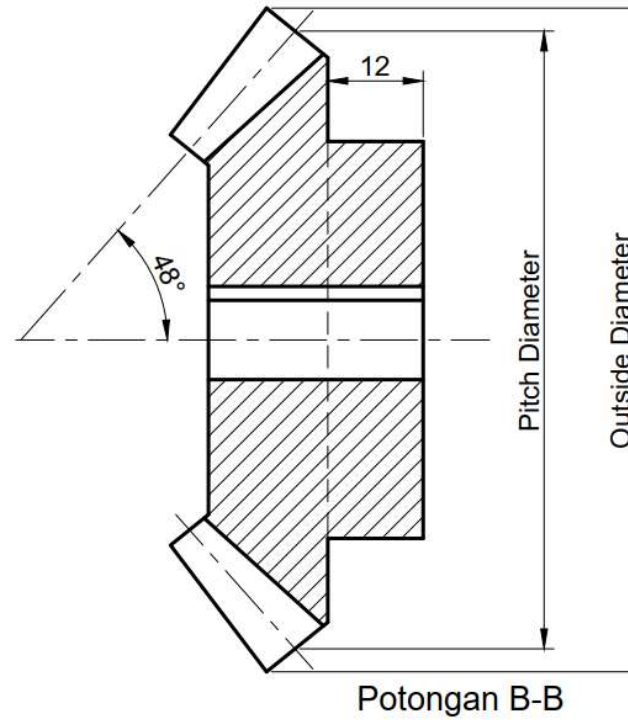
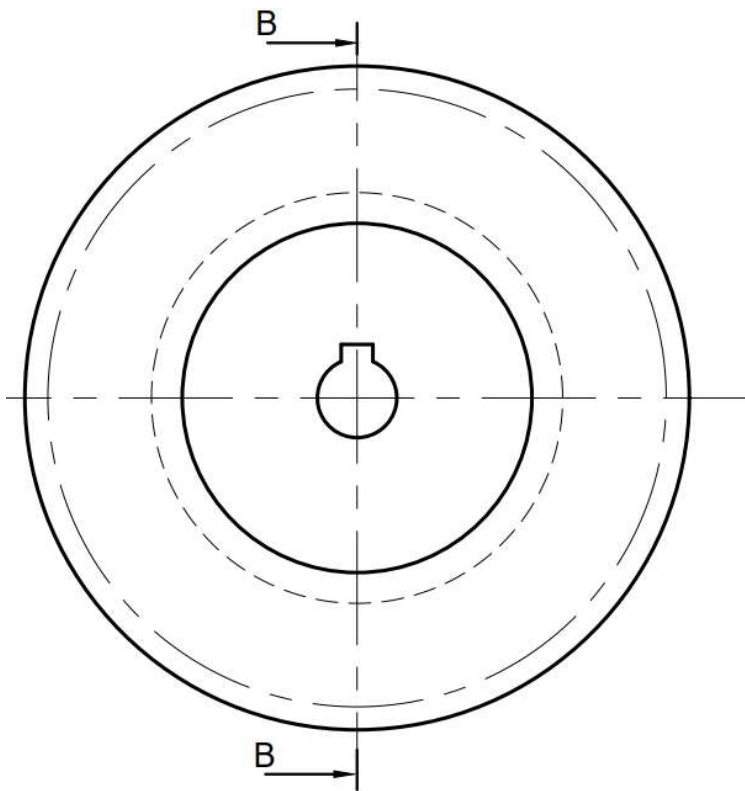
	Baut (Countersink head screw)	Tanpa skala
		Satuan: mm
	AR – Penyederhanaan Gambar	5



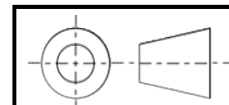
Spur gear table	
Number of teeth	18
Pressure Angle	20°
Module	2
Outside Diameter	42,4 mm
Diametral Pitch	12
Pitch Diameter	38,1 mm
Bore Size	19 mm



	Roda Gigi Lurus (Spur Gear)	Tanpa skala
		Satuan: mm
	AR – Penyederhanaan Gambar	6



Bevel gear table	
Number of teeth	20
Ratio	2:1
Outside Diameter	71,1 mm
Diametral Pitch	8
Pitch Diameter	63,5 mm
Bore Size	25,4 mm



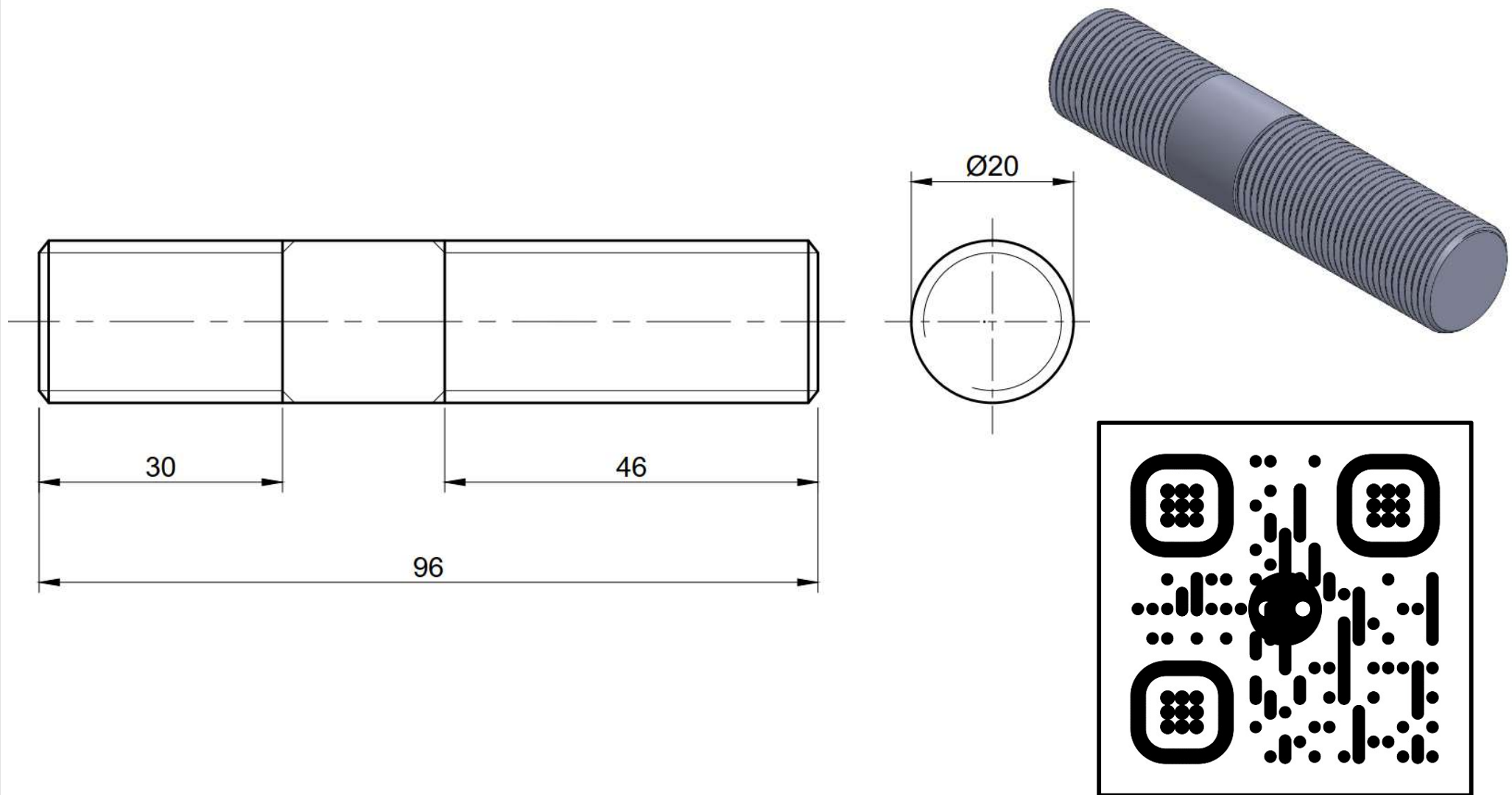
Roda Gigi Payung Lurus
(Straight Bevel Gear)

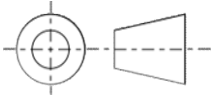
Tanpa skala

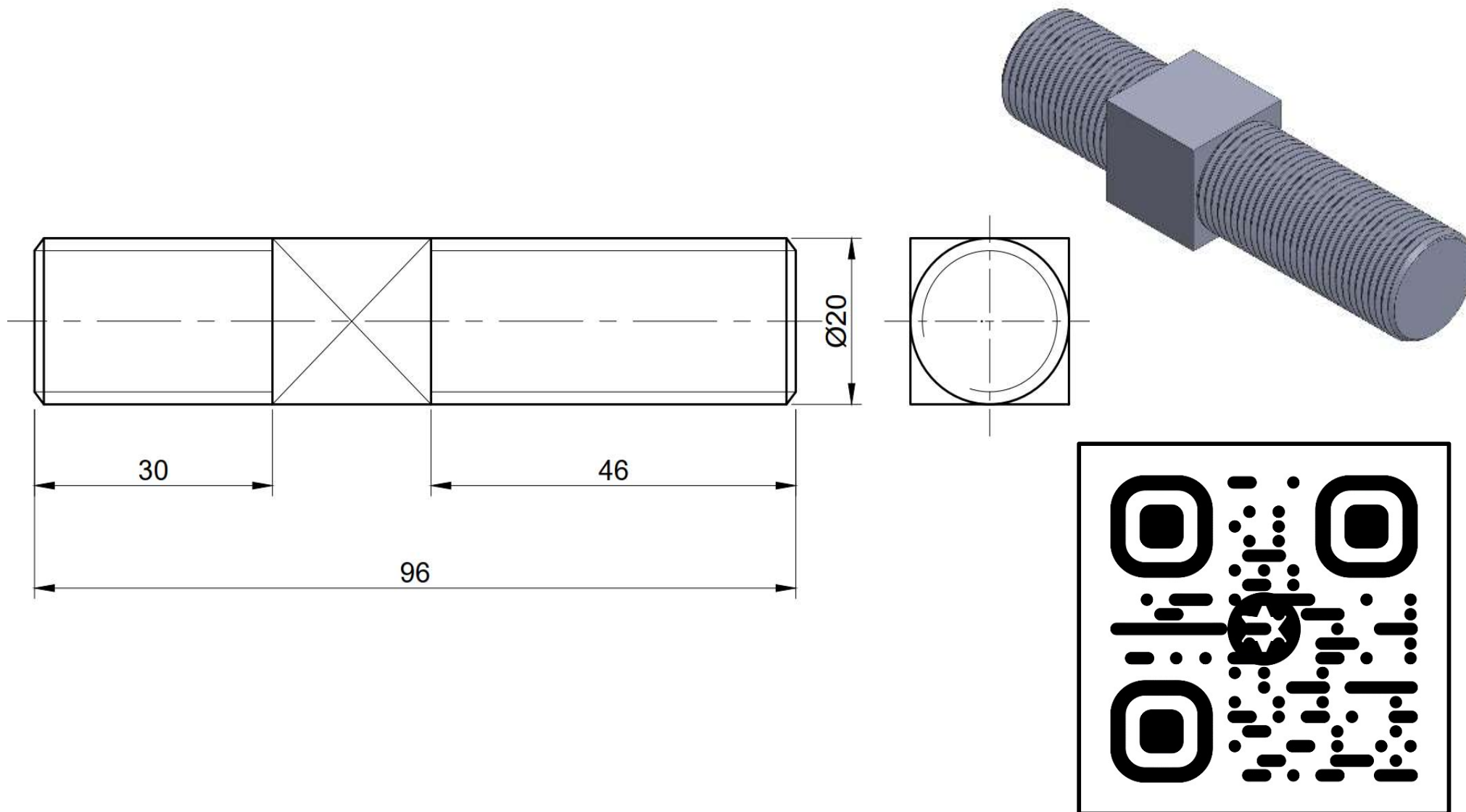
Satuan: mm

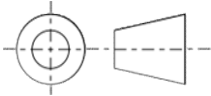
AR – Penyederhanaan Gambar

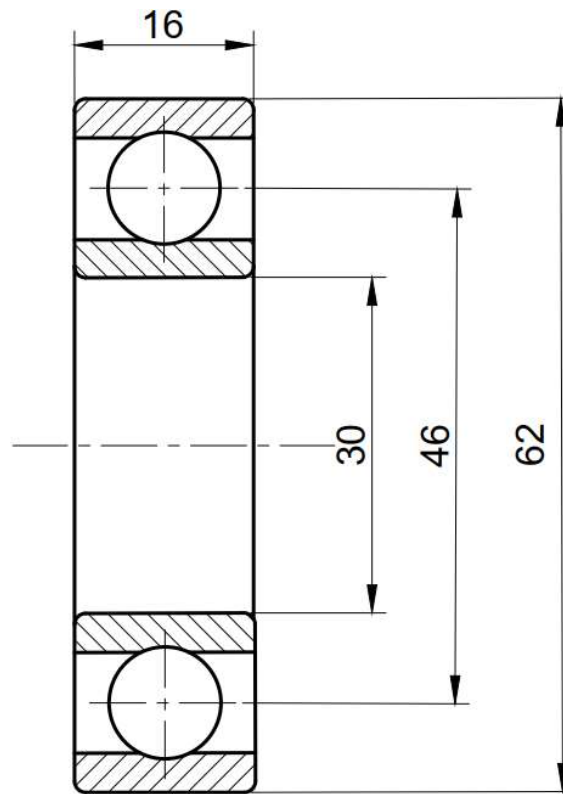
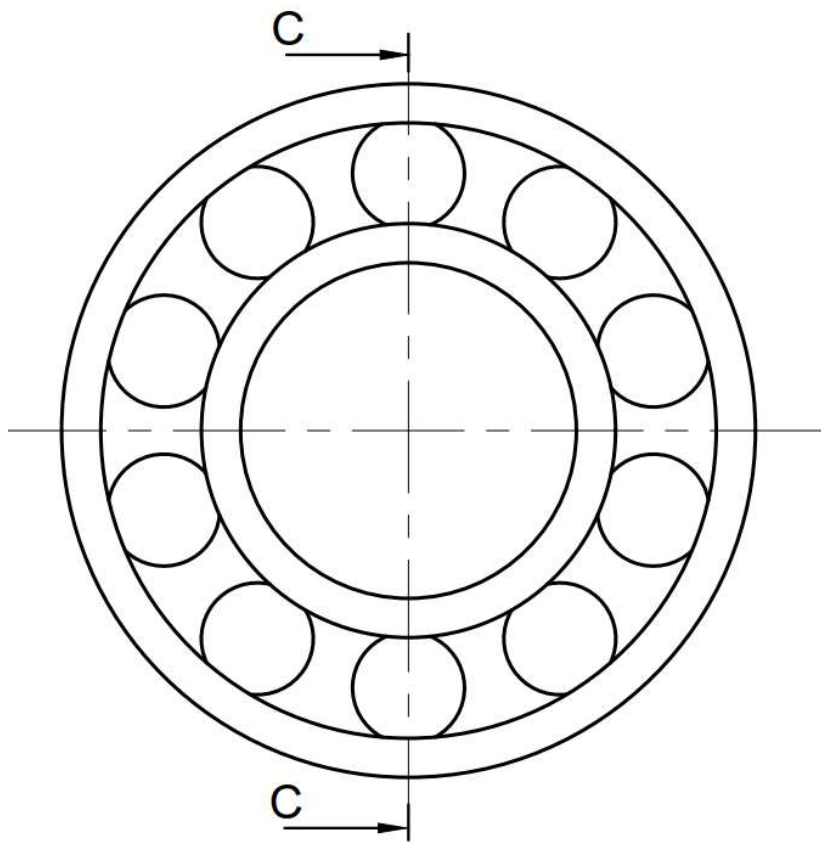
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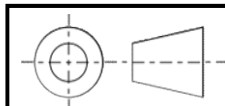
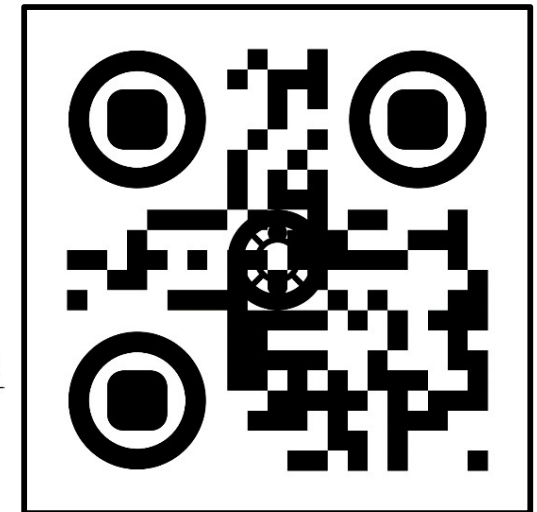
	Stud	Tanpa skala
		Satuan: mm
	AR – Penyederhanaan Gambar	8



	Boxed Stud	Tanpa skala
		Satuan: mm
	AR – Penyederhanaan Gambar	9



Potongan C-C



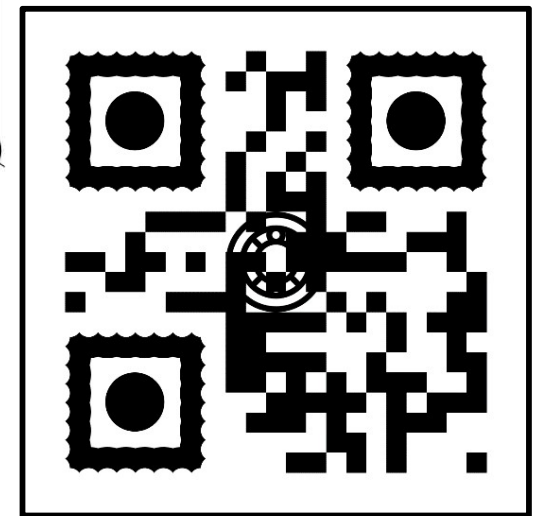
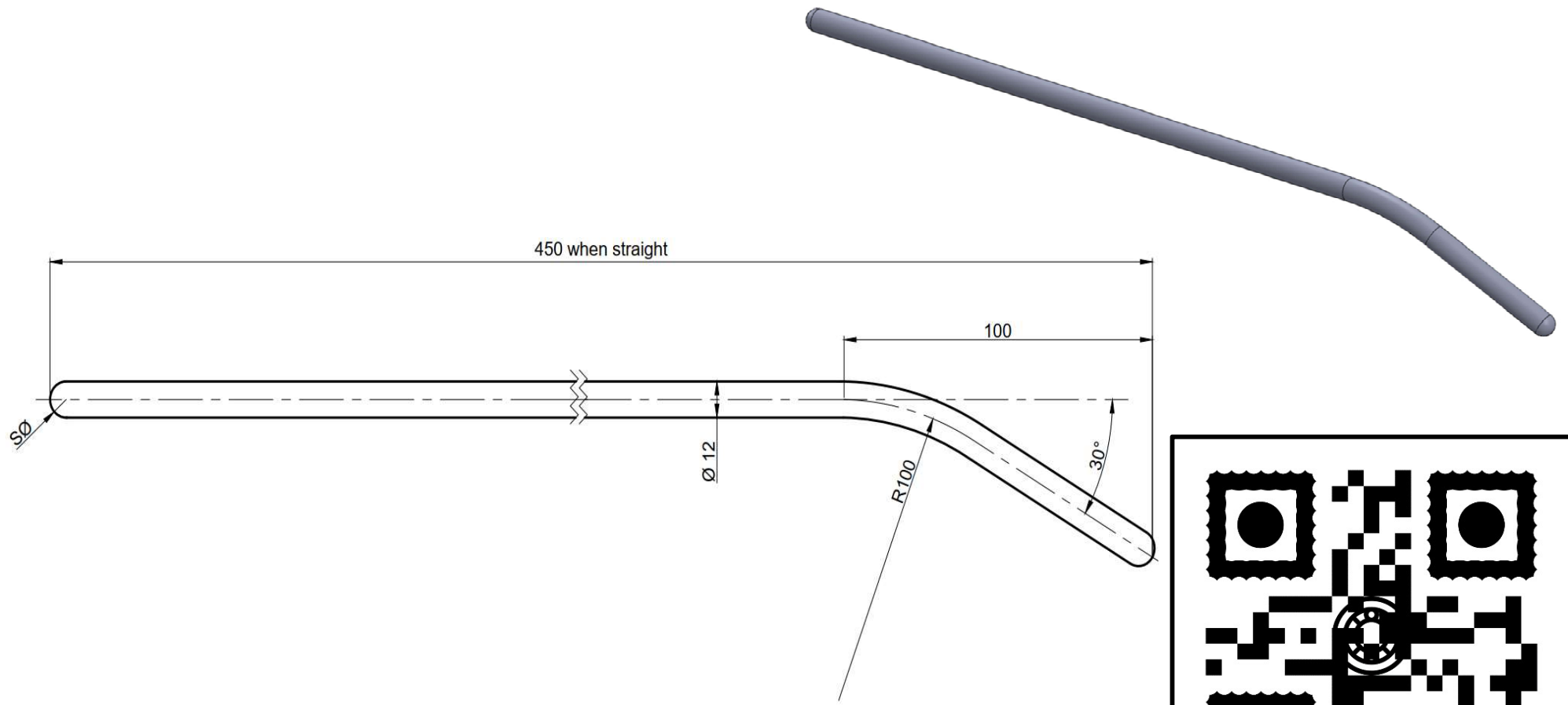
Bantalan (Bearing)

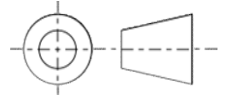
AR – Penyederhanaan Gambar

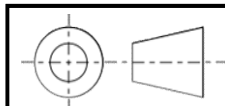
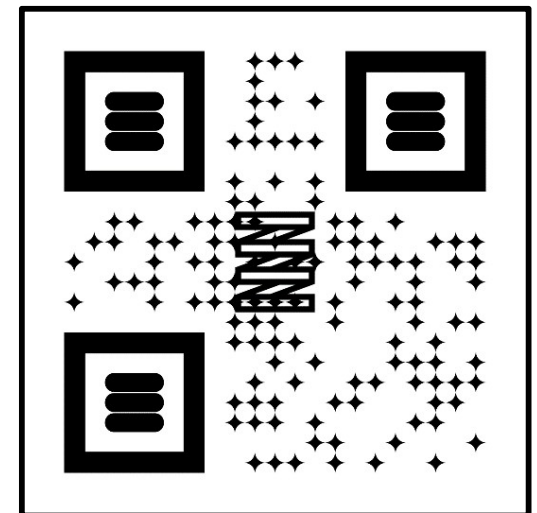
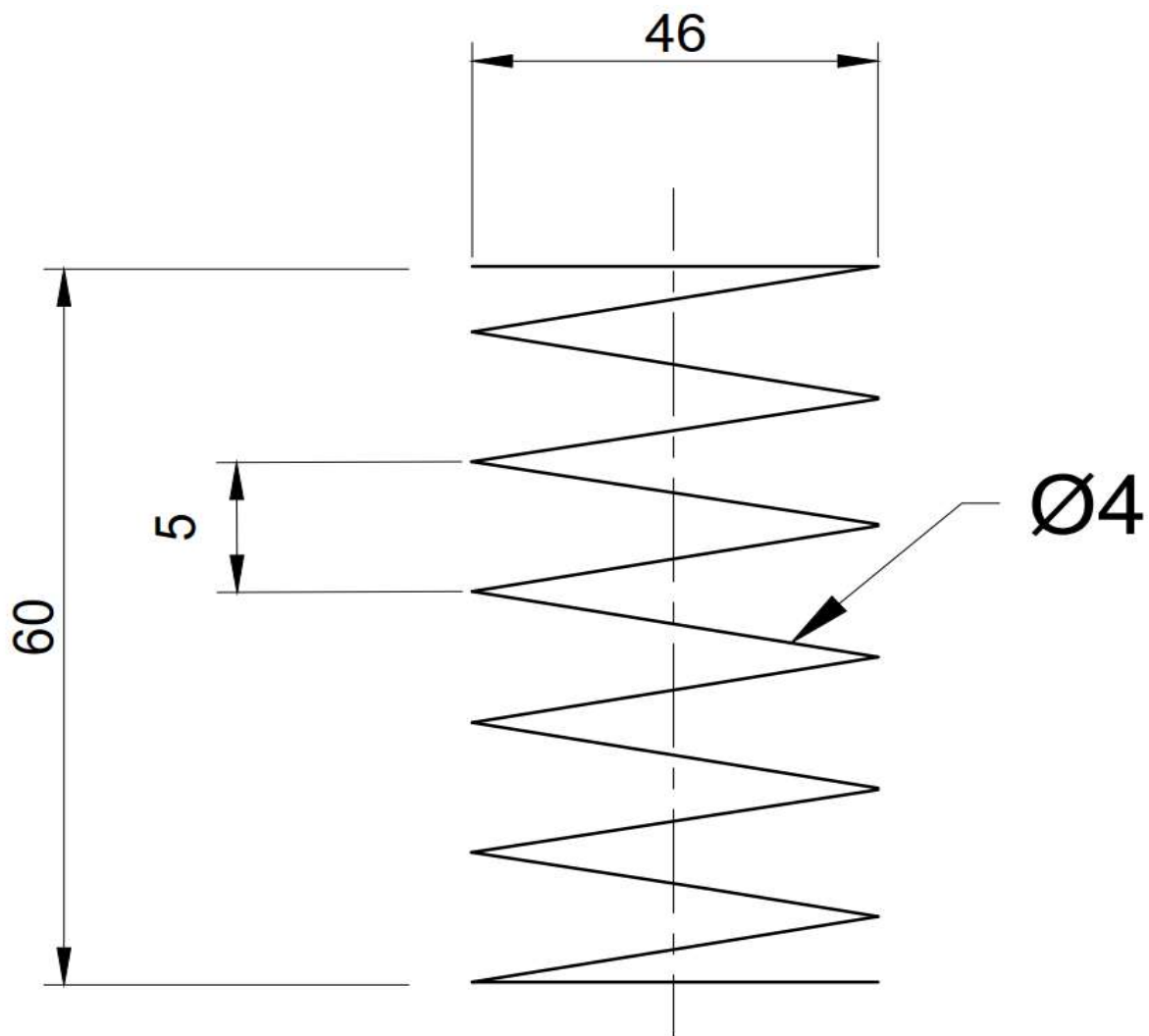
Tanpa skala

Satuan: mm

10



	Rod		Tanpa skala
			Satuan: mm
	AR – Penyederhanaan Gambar		11



Baut (Straight spring)

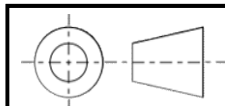
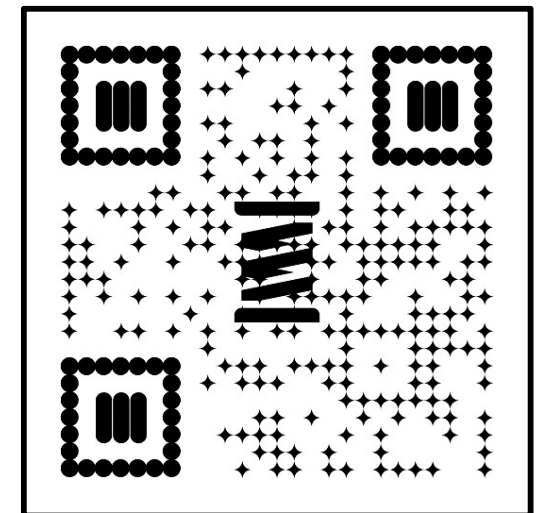
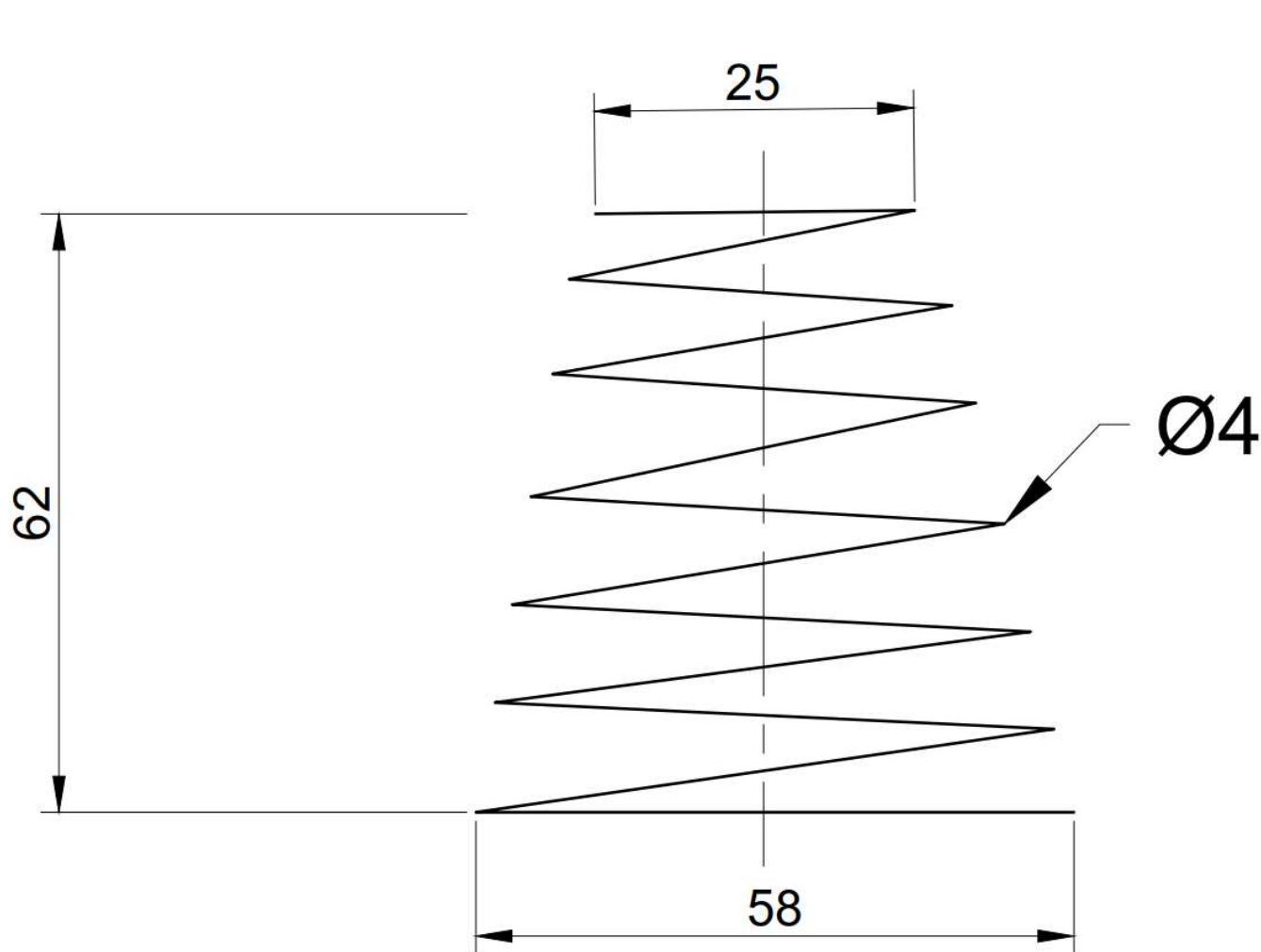
Tanpa skala

Satuan: mm



AR – Penyederhanaan Gambar

12



Pegas kerucut (Cone spring)

Tanpa skala

Satuan: mm



AR – Penyederhanaan Gambar

13

Terima Kasih

Terima kasih disampaikan kepada Direktorat Pendidikan Vokasi dan Profesi atas dukungan pada pembuatan Modul Daring Praktikum Penyederhanaan Gambar yang berbasis Augmented Reality ini melalui Program Fasilitasi Modul Pembelajaran Daring pada Pendidikan Tinggi Vokasi Tahun 2021.

Juga kepada ITS, Fakultas Vokasi ITS dan Departemen Teknik Mesin Industri ITS atas kesempatan dan dukungan dalam proses pembuatan Aplikasi AR Penyederhanaan Gambar ini.

Semoga pembuatan layout sebagai pendukung Aplikasi AR Penyederhanaan Gambar ini bermanfaat bagi mahasiswa dari berbagai perguruan tinggi dan insan pembelajar yang sedang belajar gambar teknik.

